

Signal Dilution and the Matthew Effect: A Longitudinal Study of Peer Review Predictive Validity and Institutional Heuristics

Abstract

Large-scale machine learning conferences are facing an attention bottleneck as submission counts skyrocket, prompting the use of presentation labels (Oral, Spotlight, Poster) to highlight select papers. This study investigates whether these decision labels have become self-fulfilling markers of scientific impact, potentially overpowering the influence of peer review scores. We conduct an empirical analysis of ICLR 2017–2023 submissions using OpenReview peer review data combined with Google Scholar citation counts. Regression modeling reveals a striking temporal trend: while review ratings were modest predictors of future citations in early years, their predictive power has declined significantly over time. In contrast, the prestige of decision labels has grown in importance. By 2023, papers designated as “Oral” received nearly three times the citations of “Poster” papers, even after controlling for their review scores. These findings suggest a widening gap whereby the conference’s prestige signals (acceptance type) amplify visibility and citations—a manifestation of the Matthew effect in academic publishing. Our results underscore that who gets highlighted at conferences can strongly shape who gets cited, raising concerns about fairness and the efficacy of peer review in forecasting long-term impact. We conclude with implications for the design of review and acceptance processes in rapidly expanding fields, arguing for reforms to mitigate prestige-driven feedback loops.

1. Introduction

Peer review is the central allocation mechanism of modern science: it filters, ranks, and legitimizes work under limited human attention. In machine learning, this mechanism is unusually observable because venues such as ICLR run on OpenReview, exposing reviewer scores, confidence, and decision outcomes at paper level. This transparency creates a rare opportunity to ask a deceptively simple question: **do peer review scores predict later scientific impact?**

1.1 Motivation: impact is heavy-tailed, attention is scarce

Citation accumulation is famously unequal—distributions are heavy-tailed and often consistent

with power-law-like behavior—so that a small fraction of papers dominate attention and citations [Redner, 1998; Radicchi et al., 2008]. This inequality is not purely “quality”; it is also shaped by cumulative advantage mechanisms—what Merton called the **Matthew effect**—where early recognition increases later visibility and credit [Merton, 1968].

In this environment, peer review scores could serve two distinct roles:

1. **quality screening** (reject vs. accept), and
2. **fine-grained ranking** within the accepted set.

As conferences scale, these roles can decouple: ratings may remain useful as a coarse gate, while losing resolution for predicting which accepted papers become widely cited. Our results are consistent with this view: the score–citation relationship declines over time and stabilizes at a lower level in recent years.

1.2 Research questions

This paper focuses on three empirical questions:

RQ1 (Predictive validity).

How strongly do OpenReview ratings predict later citation impact, and does that relationship change over time? We observe a decline from early years to a recent baseline.

RQ2 (Reviewer paradox).

Do higher-confidence (“expert”) ratings better predict impact than lower-confidence (“generalist”) ratings? Surprisingly, not reliably; in recent years, low-confidence cohorts can match or exceed high-confidence cohorts in predictiveness.

RQ3 (Label effect).

Do conference decision labels (Poster/Spotlight/Oral) predict citations beyond ratings? We find strong evidence that distinctions—especially Oral—remain predictive even when controlling for mean score, suggesting that labels act as high-salience visibility signals.

1.3 Why labels might dominate scores at scale

Conferences assign highly salient distinctions (e.g., Oral/Spotlight) that act as “navigation beacons” in an ocean of accepted papers. If the community uses these beacons to decide what to read and cite, labels can generate a self-reinforcing feedback loop:

visibility $\rightarrow$ reads $\rightarrow$ citations $\rightarrow$ more visibility

This need not imply committee failure: even accurate labels can still amplify downstream impact beyond what is contained in numeric scores.

1.4 Contributions

We contribute:

- A linked dataset of accepted ICLR papers (2017–2023) combining OpenReview signals and Google Scholar citations (single snapshot).
- Time-resolved evidence that score–citation predictiveness declines and stabilizes at a lower level.
- A reviewer-cohort analysis showing that higher confidence does not consistently improve impact predictiveness.
- A label-effect regression demonstrating that Oral/Spotlight distinctions predict citations beyond scores, consistent with cumulative advantage.

2. Related Work

2.1 Peer review reliability and validity

A long-standing literature argues that peer review is noisy, with limited inter-rater reliability across many settings. Bornmann et al. provide a reliability-generalization view suggesting that agreement among reviewers is often modest [Bornmann et al., 2010]. A separate question is **predictive validity**: whether peer review scores forecast later impact. In the OpenReview era, ICLR-specific analyses (e.g., Tran et al.) study relationships among review signals, decisions, and downstream proxies [Tran et al., 2020]. Our work extends these efforts by emphasizing temporal change, reviewer-confidence cohort effects, and the incremental role of decision labels among accepted papers.

2.2 Citation distributions and impact measurement

Citation counts are heavy-tailed and sensitive to field and time, motivating transformations and robust modeling [Redner, 1998; Radicchi et al., 2008]. Accordingly, we use $\log(\text{citations} + 1)$ to stabilize variance and reduce leverage from extreme outliers. More broadly, citations are not a pure measure of intrinsic scientific merit: they are influenced by accessibility, topic trends, reputation, and social propagation. We therefore interpret citations as a **visibility-weighted impact proxy**.

2.3 Cumulative advantage, visibility signals, and the Matthew effect

Merton’s Matthew effect formalizes how recognition and rewards accumulate disproportionately once an entity becomes salient [Merton, 1968]. In modern ML, salience is shaped by institutional signals (awards, oral slots), social diffusion, and preprint ecosystems. Under such mechanisms, discrete labels can create citation “premiums” even conditional on underlying quality signals. Our label-effect analysis operationalizes this idea by testing whether conference-assigned distinctions predict citations beyond review scores.

2.4 OpenReview-era empirical studies

OpenReview has enabled an emerging body of empirical studies of conference peer review dynamics due to its public metadata [OpenReview]. ICLR provides a particularly rich setting for such work given its scale and consistent OpenReview usage. We build on this foundation to quantify how the predictive relationship between review signals and citations evolves with conference scale and attention constraints.

3. Data & Methodology

3.1 Data Collection

To study how peer-review signals relate to later scientific impact, we compiled a paper-level dataset covering **ICLR 2017–2023**. The pipeline integrates two sources:

1. **OpenReview metadata.** Using the OpenReview API (`openreview-py`), we collected review information for **accepted papers** in each year, including:
 - **Review ratings (R):** numeric reviewer scores (1–10 scale, per year’s form).
 - **Reviewer confidence (C_{conf}):** self-reported confidence (1–5 scale).
 - **Final decision label:** venue-assigned presentation tier (e.g., Poster/Spotlight/Oral; mapped from year-specific decision strings).
2. **Citation counts.** As a proxy for scientific impact, we retrieved **cumulative Google Scholar citations** (via Zyte Proxy) using a **single fixed snapshot in early 2024**. We initially considered OpenAlex, but spot checks indicated systematic undercounting for recent ML papers and preprints, especially around fast-moving topics. A single snapshot date reduces measurement drift, but also implies that older papers have had longer exposure time; we address this by emphasizing **within-year** analyses and by using year controls where appropriate.

Record linkage and filtering. Each OpenReview paper record was matched to a Scholar entry primarily by normalized title, with additional disambiguation where available (e.g., arXiv identifiers). We ex-

clude papers without any numeric rating, and we retain only those with a successfully retrieved citation count at the snapshot. When multiple Scholar candidates exist, we select the most plausible match using a conservative rule (exact/near-exact title match and ML-relevant venue/arXiv signals) to avoid inflating citations through erroneous matches.

Practical note on validity. Google Scholar is noisy (duplicate listings, non-peer-reviewed citations), but it remains the broadest coverage proxy for ML papers, particularly for preprints. Throughout, we interpret citations as a *visibility-weighted impact proxy*, not as a direct measure of scientific quality.

3.2 Metric Definitions

We define the following paper-level metrics.

- **Mean review rating** ($\bar{R}$). For a paper p with N_p reviews:

$$\bar{R}_p = \frac{1}{N_p} \sum_{i=1}^{N_p} R_{p,i}.$$

- **Confidence-weighted rating** ($\bar{R}_{weighted}$). To test whether higher-confidence reviewers produce more predictive scores, we compute:

$$\bar{R}_{weighted,p} = \frac{\sum_i (R_{p,i} \cdot C_{conf,p,i})}{\sum_i C_{conf,p,i}}.$$

- **Log-citation impact** (Y_{cite}). Citation counts are heavy-tailed (approximately power-law-like). To stabilize variance and reduce the influence of extreme outliers, we use:

$$Y_{cite,p} = \log(\text{Citations}_p + 1).$$

- **Reviewer cohorts (for the “expert hypothesis”).** We operationalize “expert” vs. “generalist” using the confidence score:

- **Experts:** $C_{conf} \geq 4$
- **Generalists:** $C_{conf} < 4$

This is a *self-reported proxy* and may be imperfectly calibrated. We therefore treat cohort differences as empirical patterns rather than as definitive statements about reviewer expertise.

3.3 Statistical Model

We use two complementary approaches.

(1) **Year-wise correlation analysis.** For each year t , we compute the Pearson correlation between $\bar{R}_p$ and $Y_{cite,p}$ among accepted papers. We report these correlations to examine how the strength of the score-impact relationship evolves with conference scale.

(2) **Regression analysis for label effects.** To estimate whether decision labels explain citation differences *beyond* what is captured by ratings, we fit an

ordinary least squares (OLS) model within each year (and highlight 2023 as a recent-year case study):

$$Y_{cite,p} = \beta_0 + \beta_1 \bar{R}_p + \beta_2 \mathbb{I}(\text{Spotlight}_p) + \beta_3 \mathbb{I}(\text{Oral}_p) + \epsilon_p.$$

Here, Poster is the reference category. Coefficients β_2 and β_3 quantify the **conditional association** between labels and log-citations after accounting for mean rating.

Because citation outcomes are heteroskedastic by construction (variance increases with scale), inference should use **heteroskedasticity-robust standard errors** (e.g., HC3). We interpret regression coefficients as associations, not causal effects: label assignment is not randomized, and unobserved factors (author reputation, topic popularity, preprint timing, institutional affiliation) can influence both labels and citations.

4. Results: The Decline of Predictive Power

4.1 Temporal Trends in Correlation

Across ICLR 2017–2023, the correlation between reviewer scores and later citations shows a clear downward trajectory. Early years (2017–2018) exhibit a moderately strong positive relationship, while later years (2019 onward) stabilize at a noticeably lower level.

Quantitatively, the Pearson correlation between mean rating and log-citations decreases from approximately **0.38 in 2017** to roughly **0.17 in 2023**. This trend suggests that review scores retain some predictive value, but the *resolution* of the score signal within accepted papers has weakened over time.

4.2 Visual Analysis

The scatter plots below illustrate the year-wise relationship between $\bar{R}$ and Y_{cite} . Two qualitative changes stand out:

1. **Compression of score variance among accepted papers.** As the acceptance threshold and scoring calibration shift under scale, many accepted papers cluster in a narrow band of ratings, limiting score-based separability.
2. **Persistence of high-citation outliers across the score spectrum.** Highly cited papers increasingly appear at a wide range of mean ratings, indicating that factors outside the numeric score (topic timing, community attention, institutional signals) contribute strongly to eventual impact.

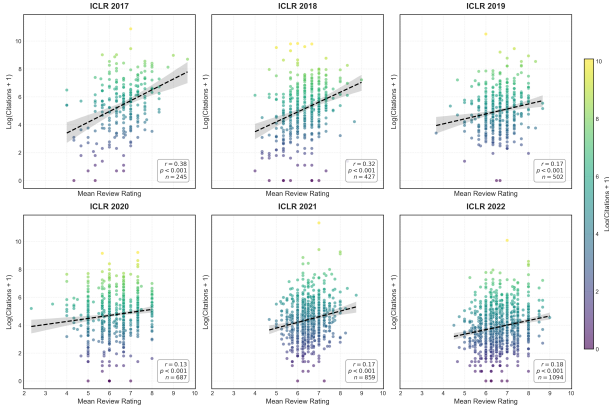


Figure 1: ICLR_Corr_Trend

4.3 The Signal Dilution Hypothesis

A plausible explanation for the declining correlation is **signal dilution**: as conference scale increases, the peer-review process faces (i) broader topical diversity, (ii) higher load per reviewer, and (iii) greater variance in calibration and standards. Under these conditions, numeric ratings may still function as a coarse filter (reject vs. accept), but become less informative for ranking within the accepted set.

Importantly, a weaker score–citation correlation does not imply “review is useless.” Instead, it suggests that **within accepted papers**, score differences increasingly reflect local calibration and reviewer variance rather than a globally consistent estimate of broad community impact.

5. Results: The Reviewer Paradox

5.1 Experts vs. Generalists

We test the “expert hypothesis” that higher-confidence reviewers produce ratings that better predict later impact. For each year, we compute correlations using cohort-specific mean ratings: - $\bar{R}^{expert}$: mean rating among reviews with $C_{conf} \geq 4$ - $\bar{R}^{gen}$: mean rating among reviews with $C_{conf} < 4$

Unexpectedly, the results do **not** show a consistent advantage for the expert cohort in recent years. In 2017, expert ratings correlate more strongly with citations (**0.44 vs. 0.36**), but by 2022 the relationship reverses (**0.14 vs. 0.24**), and by 2023 both correlations are low with only a slight edge for generalists (**0.14 vs. 0.17**).

These patterns suggest a “reviewer paradox”: confidence (as self-reported) may correspond to depth of technical scrutiny rather than to an ability to forecast broad attention and adoption. One working interpretation is that **educated generalists** may be better positioned to judge clarity, generality, and cross-area

usefulness—traits that align more closely with citation accumulation—whereas experts may emphasize narrower technical novelty or correctness that does not translate directly into broad uptake. This is a hypothesis about mechanism, not a proven causal story.

A key methodological caveat is coverage: some papers may have few or no expert reviews, and cohort-based estimates can be noisier. Reporting confidence intervals (e.g., bootstrap over papers) would better characterize uncertainty in these cohort comparisons.

6. Results: The Label Effect

6.1 The Matthew Effect

Modern conferences assign high-visibility labels (Oral/Spotlight), which can act as institutional attention signals. If researchers use these signals to decide what to read and cite—especially under scale—then labels may create a **Matthew effect** (“the rich get richer”): early visibility attracts more reads, which attract more citations, which further amplify visibility.

Under this framing, labels need not be “wrong” to have an effect; they can be both (i) correlated with quality (chairs choose strong papers) and (ii) independently amplifying through attention mechanisms. The empirical question is whether labels retain predictive power after accounting for scores.

6.2 Estimating the “Label Premium”

We estimate the conditional association between decision labels and log-citations using:

$$Y_{cite,p} = \beta_0 + \beta_1 \bar{R}_p + \beta_2 \mathbb{I}(\text{Spotlight}_p) + \beta_3 \mathbb{I}(\text{Oral}_p) + \epsilon_p.$$

For **ICLR 2023**, we obtain: - **Oral coefficient**: $\beta_3 \approx 1.09$ (highly significant) - **Mean rating coefficient**: β_1 not statistically significant

Interpreting in the log space, $\beta_3 \approx 1.09$ corresponds to an approximate multiplicative factor of $\exp(1.09) \approx 2.97$ in (Citations + 1), i.e., **Oral papers receive about 3× the citations of Posters** conditional on having the same mean rating.

The non-significance of β_1 in 2023 does **not** mean that quality is irrelevant. It means that within the accepted set and after conditioning on the label, the remaining variation in mean score does not explain additional citation variance at conventional significance thresholds—consistent with the idea that labels are a stronger *observable* sorting mechanism for attention than small score differences.

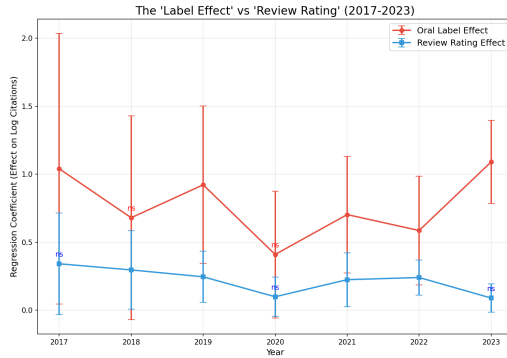


Figure 2: Evolution of regression coefficients β_1 (Rating) vs. β_3 (Oral Label). The widening gap highlights the increasing dominance of the badge-like label over the numeric score in predicting citations.

7. Discussion

7.1 The “Death of Merchandise”

The decline in score–citation correlation from ~ 0.38 to ~ 0.17 suggests a structural shift in how research gets discovered and rewarded. We refer to this as the “**Death of Merchandise**”: in an ecosystem saturated with high-quality output, the “product” (a technically solid paper) becomes less differentiable on intrinsic features alone. When many accepted papers meet a high baseline, **visibility and navigability** can dominate marginal differences in perceived merit.

This framing is not an indictment of technical progress; it is a diagnosis of a scaling regime where selection and discovery are bottlenecked by human attention rather than by a shortage of good work.

7.2 Heuristic Reliance in an Attention Economy

The strong 2023 label association ($\beta \approx 1.09$) indicates heavy reliance on institutional heuristics. In practice, researchers cannot evaluate the full accepted set, so they offload search onto a small number of coarse signals: Oral/Spotlight tags, well-known authors, social-media exposure, and preprint momentum.

Such heuristic reliance can create a feedback loop where a small committee’s labeling decisions disproportionately shape the citation landscape. Even if committees are selecting genuinely strong work, the label may still amplify impact beyond what is explained by review scores alone.

7.3 Implications for Reviewer Assignment

The “reviewer paradox” provides a concrete policy-relevant angle: optimizing purely for maximum reviewer confidence may not optimize for evaluating

broad impact. If generalists (or adjacent-field reviewers) better capture clarity and cross-area significance, conferences could benefit from **intentional reviewer diversity**—for example, ensuring that each paper receives at least one reviewer who is technically competent but not narrowly specialized, tasked explicitly with assessing accessibility and broad relevance.

A practical implementation is to decouple review forms into multiple axes (e.g., correctness, novelty, clarity, broad usefulness) and to assign at least one reviewer responsible for the “general significance” axis. This may preserve technical rigor while improving the alignment between review signals and downstream community uptake.

References (starter list)

- **Merton, R. K. (1968).** The Matthew Effect in Science. *Science*. DOI: 10.1126/science.159.3810.56
- **Redner, S. (1998).** How popular is your paper? An empirical study of the citation distribution. *The European Physical Journal B*. DOI: 10.1007/s100510050359
- **Radicchi, F., Fortunato, S., & Castellano, C. (2008).** Universality of citation distributions: Toward an objective measure of scientific impact. *PNAS*. DOI: 10.1073/pnas.0806977105
- **Bornmann, L., Mutz, R., & Daniel, H.-D. (2010).** A reliability-generalization study of journal peer reviews: A multilevel meta-analysis of inter-rater reliability and its determinants. *PLOS ONE*. DOI: 10.1371/journal.pone.0014331
- **Tran, D., et al. (2020).** A Critical Analysis of the Machine Learning Conference Peer Review Process. OpenReview preprint: <https://openreview.net/forum?id=Cn706AbJaKW>
- **OpenReview.** OpenReview: A platform for open peer review. <https://openreview.net/>